# Homework Exercise Sheet

## Ex. 3 Solution

Label (S) : only one correct answer; (M) : more than one correct answer

1. (S) Select the linear system of equations that seeks the solution $\mathbf{x}^*$ of the following least-squares problem

$$\mathbf{x}^* = \underset{\mathbf{x}\in\mathbb{R}^n}{\operatorname{argmin}} ||\mathbf{A}\mathbf{x} - \mathbf{b}||_2^2 + \lambda||\mathbf{x}||_2^2$$

where $\mathbf{A} \in \mathbb{R}^{m,n}, \mathbf{b} \in \mathbb{R}^m, m > n$.

a) $(\mathbf{A}^\top\mathbf{A} + 2\lambda\mathbf{I})\mathbf{x}^* = \mathbf{A}^\top\mathbf{b}$.

b) $(\mathbf{A}^\top\mathbf{A} - 2\lambda\mathbf{I})\mathbf{x}^* = \mathbf{A}^\top\mathbf{b}$.

c) $(\mathbf{A}^\top\mathbf{A} + \lambda\mathbf{I})\mathbf{x}^* = \mathbf{A}^\top\mathbf{b}$.

d) $(\mathbf{A}^\top\mathbf{A} - \lambda\mathbf{I})\mathbf{x}^* = \mathbf{A}^\top\mathbf{b}$.

**Solution:**

**c).** The problem seeks the minimizer of $J(\mathbf{x}) = \mathbf{x}^\top\mathbf{A}^\top\mathbf{A}\mathbf{x} - \mathbf{2b}^\top\mathbf{A}\mathbf{x} + \mathbf{b}^\top\mathbf{b} + \lambda\mathbf{x}^\top\mathbf{x}$. Thus, $\mathbf{x} \in \mathbb{R}^n$ is a minimizer only if $\mathbf{grad}J(\mathbf{x}) = 2\mathbf{A}^\top\mathbf{A}\mathbf{x} - \mathbf{2A}^\top\mathbf{b} + \mathbf{2}\lambda\mathbf{x} = 0$.

2. (M) Select cases that have a **unique** solution for the following least-squares problem

$$\mathbf{x}^* = \underset{\mathbf{x}\in\mathbb{R}^n}{\operatorname{argmin}} ||\mathbf{A}\mathbf{x} - \mathbf{b}||_2^2.$$

a) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$

b) $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$

c) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 4 & 8 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$

d) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 5 & 6 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 7 \\ 0 \\ 9 \end{bmatrix}$

**Solution:**

**a), d).** According to Corollary 3.1.2.13 in the lecture document, uniqueness can be checked by testing if $m \geq n$ and $\operatorname{rank}(\mathbf{A}) = n$. Here, $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

(a) Unique: $3 = m \geq n = 2$, $\operatorname{rank}(\mathbf{A}) = 2 = n$.

(b) Not unique: $2 = m < n = 3$.

(c) Not unique: $3 = m \geq n = 2$, $\operatorname{rank}(\mathbf{A}) = 1$.

(d) Unique: $3 = m \geq n = 2$ and $\operatorname{rank}(\mathbf{A}) = 2 = n$.

3. (S) Consider the following least-squares problem:

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2, \mathbf{A} = \begin{bmatrix} 1 & 2 & -1 & -4 \\ 2 & 4 & -2 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 4 & 16 \\ 1 & 2 & 3 & 12 \\ -1 & -2 & 2 & 8 \end{bmatrix}, \mathbf{x}, \mathbf{b} \in \mathbb{R}^4.$$

What is the dimension of the set of all $\mathbf{x}^*$ that minimize the problem?

a) 1 b) 2 c) 3 d) 4

**Solution:**

**b).** The dimension of the solution set is determined by the dimension of the null space of $\mathbf{A}^\intercal \mathbf{A}$ in the normal equation. As theorem 3.1.2.9 shows, $\mathcal{N}(\mathbf{A}^\intercal \mathbf{A}) = \mathcal{N}(\mathbf{A})$. Since $\mathbf{A}$ is of rank 2, due to the fact that the first two and the last two columns of $\mathbf{A}$ are linearly dependent, the kernel space $\mathcal{N}(\mathbf{A})$ is two-dimensional.

4. (M) Consider $\mathbf{A} \in \mathbb{R}^{m,n}, \mathbf{b} \in \mathbb{R}^m$ where $\mathbf{A}$ is **full rank**. An "exact" solution $\mathbf{x} \in \mathbb{R}^n$ is defined to be satisfying the linear system of equations $\mathbf{A}\mathbf{x} = \mathbf{b}$ exactly. Which of the following statements regarding existence and uniqueness of exact solutions are **true**?

a) If $m = n$, there always exists a unique exact solution.

b) If $m < n$, there always exist more than one exact solution.

c) If $m > n$, there may exist exact solutions.

**Solution:**

**a), b), c).** First we discuss existence, which boils down to the question of "whether $\mathbf{b}$ belongs to the range space $\mathcal{R}(\mathbf{A})$". Since $\mathbf{A}$ is full rank, it holds that $\mathcal{R}(\mathbf{A}) = \mathbb{R}^m$ when $m \leq n$. It leads to existence of exact solution in the case $m \leq n$. When $m > n$, $\mathcal{R}(\mathbf{A})$ is a strict subset of $\mathbb{R}^m$, which does not always admit a solution. Concerning uniqueness, it is about "whether the kernel space $\mathcal{N}(\mathbf{A}) = \emptyset$". When $m < n$, it always holds that $\mathcal{N}(\mathbf{A}) \neq \emptyset$, which results in multiple solutions; when $m \geq n$, $\mathcal{N}(\mathbf{A}) = \emptyset$ (see Remark 3.1.2.15) and thus only one unique solution is admitted. Notice that all the discussion relies on the assumption that $\mathbf{A}$ is full rank.

5. (M) Assume a linear model is applicable to represent the dependency between coordinates $(x, y, z)$ and temperature $t$, that is, there exist parameters $a, b, c, d \in \mathbb{R}$ such that $t(x, y, z) = ax+by+cz+d$. The available data consists of 150 temperature measurements taken at different locations. What are the correct statements about the shapes of matrices $\mathbf{Q}$ and $\mathbf{P}$ in the **normal equations**, $\mathbf{Q}\mathbf{x} = \mathbf{P}$, used to obtain the least-squares solution $\mathbf{x}$ representing model parameters?

a) $\mathbf{Q} \in \mathbb{R}^{150,3}$ b) $\mathbf{Q} \in \mathbb{R}^{150,4}$ c) $\mathbf{Q} \in \mathbb{R}^{3,3}$ d) $\mathbf{Q} \in \mathbb{R}^{4,4}$

e) $\mathbf{Q} \in \mathbb{R}^{150,150}$ f) $\mathbf{P} \in \mathbb{R}^3$ g) $\mathbf{P} \in \mathbb{R}^4$ h) $\mathbf{P} \in \mathbb{R}^{150}$

**Solution:**

**d), g).** A single measurement defines the relation $ax_i + by_i + cz_i + d = t_i$, where $(x_i, y_i, z_i) \in \mathbb{R}^3$ represents a point in three-dimensional space, and $t_i \in \mathbb{R}$ represents a temperature. Here, $\mathbf{x} = (a, b, c, d)^\top \in \mathbb{R}^4$ are the parameters of a linear model. In this case, the over-determined linear system of equations is given by $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{150,4}$ and $\mathbf{b} \in \mathbb{R}^{150}$. The corresponding normal equations are defined as follows: $\mathbf{Q} = \mathbf{A}^\top\mathbf{A} \in \mathbb{R}^{4,4}$, and $\mathbf{P} = \mathbf{A}^\top\mathbf{b} \in \mathbb{R}^4$.

6. (S) Consider the overdetermined linear systems of equations $\mathbf{Ax} = \mathbf{b}$, with $\mathbf{A} \in \mathbb{R}^{m,n}, \mathbf{b} \in \mathbb{R}^m$ given $m > n$. By introducing the residual $\mathbf{r}^* := \mathbf{Ax}^* - \mathbf{b}$ as additional auxiliary variable the least-squares solution $\mathbf{x}^*$ can be yield by solving so-called **extended normal equations**. Please select the correct linear system of equations to be solved. *Note that* $\mathbf{I}$ *denotes identity matrix and* $\mathbf{0}$ *denotes zero matrix or vector.*

   a) $\begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^\top & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{r}^* \\ \mathbf{x}^* \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}$

   b) $\begin{bmatrix} -\mathbf{I} & \mathbf{A} \\ \mathbf{A}^\top & \mathbf{A}^\top\mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{r}^* \\ \mathbf{x}^* \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix}$

   c) $\begin{bmatrix} -\mathbf{0} & \mathbf{A} \\ \mathbf{A}^\top & \mathbf{A}^\top\mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{r}^* \\ \mathbf{x}^* \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{A}^\top\mathbf{b} \end{bmatrix}$

   d) $\begin{bmatrix} -\mathbf{I} & \mathbf{A} \\ \mathbf{A}^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{r}^* \\ \mathbf{x}^* \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}$

**Solution:**

**d).** With the standard normal equation method, we yield $\mathbf{x}^*$ by solving $\mathbf{A}^\top\mathbf{A}\mathbf{x}^* = \mathbf{A}^\top\mathbf{b} \Leftrightarrow \mathbf{A}^\top(\mathbf{A}\mathbf{x}^* - \mathbf{b}) = \mathbf{0} \Leftrightarrow \mathbf{A}^\top\mathbf{r}^* = \mathbf{0}$. Combined with the definition of the residual $\mathbf{r}^* := \mathbf{Ax}^* - \mathbf{b}$, the solution is obtained. The advantage of this approach is that, if $\mathbf{A}$ is a sparse matrix, the sparsity of $\mathbf{A}$ is preserved while $\mathbf{A}^\top\mathbf{A}$ involved in the standard normal equation method could break the sparsity.

7. (M) Below is the code to solve a linear least-squares problem via normal equations. Consider $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{b} \in \mathbb{R}^m$. Choose the correct statements about the complexity of the implementation fragments indicated with comments at line 4.

```
1 VectorXd normeqsolve(const MatrixXd &A,const VectorXd &b) {
2   if (b.size() != A.rows()) throw runtime_error("Dimension mismatch");
3   VectorXd x = (A.transpose()*A).llt().solve(A.transpose()*b);
4                //  <-------1------->  <----2----> <-------3------->
5   return x;
6 }
```

   a) Fragment 1: $O(nm^2)$.
   b) Fragment 2: $O(nm^2)$.
   c) Fragment 3: $O(nm^2)$.
   d) Fragment 1: $O(mn^2)$.
   e) Fragment 2: $O(mn^2)$.
   f) Fragment 3: $O(mn^2)$.
   g) Fragment 1: $O(mn)$.
   h) Fragment 2: $O(mn)$.
   i) Fragment 3: $O(mn)$.
   j) Fragment 1: $O(n^3)$.
   k) Fragment 2: $O(n^3)$.
   l) Fragment 3: $O(n^3)$.

**Solution:**

**d), h), l).** See §3.2.0.2 in the lecture document. Fragment 1 complexity is $O(mn^2)$. Fragment 2 complexity is $O(mn)$. Fragment 3 complexity is $O(n^3)$.

8. (S) What is the number of iterations $k$ needed to solve the least-square problem

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2, \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^n, \mathbf{b} \in \mathbb{R}^m$$

under a given error bound $\|\mathbf{x}^* - \mathbf{x}_k\| < \varepsilon$ by using second order optimization method (Newton's method)? In the second order optimization method, one start from an initial guess of the solution $\mathbf{x_0}$ and improve it iteratively according to the following formula

$$\mathbf{x}_{i+1} = \mathbf{x}_i - \mathbf{H}^{-1} \nabla J(\mathbf{x}_i), i = 0, 1, \cdots, k-1$$

where $J(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ is the objective function that we would like to minimize, $\mathbf{H} = \nabla^2 J(\mathbf{x})$ is the Hessian matrix of the objective function.

**Hint:** Check Remark 3.1.2.5 in Lecture Note for the gradient of $J$.

a) $O(1)$

b) $O(\|\mathbf{x}^* - \mathbf{x}_0\|/\varepsilon)$

c) $O(\log(\|\mathbf{x}^* - \mathbf{x}_0\|/\varepsilon))$

d) $O(\sqrt{\|\mathbf{x}^* - \mathbf{x}_0\|/\varepsilon})$

**Solution:**

a). We can expand the iteration formula by hand: $\nabla J(\mathbf{x}_i) = 2\mathbf{A}^\intercal\mathbf{A}\mathbf{x} - 2\mathbf{A}^\intercal\mathbf{b}$ and $\mathbf{H} = \nabla(\nabla J(\mathbf{x})) = 2\mathbf{A}^\intercal\mathbf{A}$. Then $\mathbf{x_1} = \mathbf{x_0} - (2\mathbf{A}^\intercal\mathbf{A})^{-1}(2\mathbf{A}^\intercal\mathbf{A}\mathbf{x}_0 - 2\mathbf{A}^\intercal\mathbf{b}) = \mathbf{x}_0 - \mathbf{x}_0 + (2\mathbf{A}^\intercal\mathbf{A})^{-1}2\mathbf{A}^\intercal\mathbf{b} = (\mathbf{A}^\intercal\mathbf{A})^{-1}\mathbf{A}^\intercal\mathbf{b}$ which is exactly the analytic solution according to the Normal Equation Method (see Lecture Note 3.2). In summary, it takes only **one iteration** for the second order optimization method to reach the analytical solution of least-square problem (which is of course within any error bounds). This statement holds for any objective functions with a quadratic form.

9. (S) What is the minimum $L_2$ distance one can get when using a 3rd order polynomial with real coeffcients $p_3(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$ to approximate the $\sin(x)$ function within range $[0, \frac{\pi}{2}]$ (truncated to 1 effective digit)?

The $L_2$ distance of two functions $f(x), g(x)$ in range $[a, b]$ is $\|f(x) - g(x)\|_{L_2} = \sqrt{\int_a^b |f(x) - g(x)|^2 dx}$.

**Hint:** Let $\mathbf{a} = (a_0, a_1, a_2, a_3)$, the problem aims to minimize

$$J(\mathbf{a}) = \|\sin(x) - p_3(x)\|_{L_2}^2 = \int_0^{\frac{\pi}{2}} (\sin(x) - \sum_{i=0}^{3} a_i x^i)^2 dx.$$

We can formulate the normal equations by setting its gradient to zeros:

$$\frac{\partial J}{\partial a_j} = \int_0^{\frac{\pi}{2}} 2(\sin(x) - \sum_{i=0}^{3} a_i x^i) x^j dx = 0, j = 0, 1, 2, 3.$$

This can be simplified to a matrix equation just like the least-square problem. Please solve that equation and insert the solution $\mathbf{a}^*$ back to $\sqrt{J(\mathbf{a})}$ to get the final value.

Here are the values of integrals needed in the process.

| | $1$ | $x$ | $x^2$ | $x^3$ | $\sin(x)$ |
|---|---|---|---|---|---|
| $1$ | $\frac{\pi}{2}$ | $\frac{\pi^2}{8}$ | $\frac{\pi^3}{24}$ | $\frac{\pi^4}{64}$ | $1$ |
| $x$ | | $\frac{\pi^3}{24}$ | $\frac{\pi^4}{64}$ | $\frac{\pi^5}{160}$ | $1$ |
| $x^2$ | | | $\frac{\pi^5}{160}$ | $\frac{\pi^6}{384}$ | $\pi - 2$ |
| $x^3$ | | | | $\frac{\pi^7}{896}$ | $\frac{3}{4}\pi^2 - 6$ |
| $\sin(x)$ | | | | | $\frac{\pi}{4}$ |

a) $10^{-3}$ b) $10^{-4}$ c) $10^{-5}$ d) $10^{-6}$

**Solution:**

**a).** Following the instructions, one can construct the normal equation as follows:

$$\sum_{i=0}^{3} a_i \int_0^{\frac{\pi}{2}} x^i x^j dx = \int_0^{\frac{\pi}{2}} \sin(x) x^j dx, j = 0, 1, 2, 3.$$

Or in matrix format $\mathbf{M}\mathbf{a}^\mathsf{T} = \mathbf{N}$:

$$\begin{bmatrix} \frac{\pi}{2} & \frac{\pi^2}{8} & \frac{\pi^3}{24} & \frac{\pi^4}{64} \\ \frac{\pi^2}{8} & \frac{\pi^3}{24} & \frac{\pi^4}{64} & \frac{\pi^5}{160} \\ \frac{\pi^3}{24} & \frac{\pi^4}{64} & \frac{\pi^5}{160} & \frac{\pi^6}{384} \\ \frac{\pi^4}{64} & \frac{\pi^5}{160} & \frac{\pi^6}{384} & \frac{\pi^7}{896} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \pi - 2 \\ \frac{3}{4}\pi^2 - 6 \end{bmatrix}.$$

Then we can evaluate those symbolic expressions in the equation at 64-bit floating point number (with approximately 15 effective decimal digits), and solve the equation numerically (using LU decomposition or other techniques). It is also possible to solve it symbolically. The result is:

$$a_0^* = -0.00225843680833363, a_1^* = 1.02716934420496,$$
$$a_2^* = -0.0699434191879021, a_3^* = -0.113868609932294$$

To substitute it in the expression of $J(\mathbf{a}^*)$ with best accuracy, we need to re-orgainze the formula and factor out all the integrals:

$$\begin{aligned} J(\mathbf{a}^*) &= \int_0^{\frac{\pi}{2}} (\sin(x) - \sum_{i=0}^{3} a_i^* x^i)^2 dx = \int_0^{\frac{\pi}{2}} \sin^2(x) - 2\sum_{i=0}^{3} a_i^* x^i \sin(x) + (\sum_{i=0}^{3} a_i^* x^i)^2 dx \\ &= \int_0^{\frac{\pi}{2}} \sin^2(x) dx - 2\sum_{i=0}^{3} a_i^* \int_0^{\frac{\pi}{2}} x^i \sin(x) + \sum_{j=0}^{3} a_j^* \sum_{i=0}^{3} a_i^* \int_0^{\frac{\pi}{2}} x^i x^j dx \\ &= \frac{\pi}{4} - 2\mathbf{a}^*\mathbf{N} + \mathbf{a}^*\mathbf{M}\mathbf{a}^{*\mathsf{T}} \\ &= \frac{\pi}{4} - \mathbf{a}^*\mathbf{N} \qquad (\text{Substitute with } \mathbf{M}\mathbf{a}^{*\mathsf{T}} = \mathbf{N}) \\ &= \frac{\pi}{4} - a_0^* - a_1^* - (\pi - 2)a_2^* - (\frac{3}{4}\pi^2 - 6)a_4^* \\ &\approx 1.09021919224818 \cdot 10^{-6}. \end{aligned}$$

In the end, the minimum $L_2$ distance is given by $\sqrt{J(\mathbf{a}^*)} \approx 1.04413561966259 \cdot 10^{-3}$.